

Capstone SS2

Part A

Client: STADIOEquities

Related work and Dataset Description 1

Title	A data-driven approach to predict the success of bank telemarketing
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', Decision Support Systems, 62, pp. 22–31.
Link	https://doi.org/10.1016/j.dss.2014.03.001
Modelling	Compared four classifiers: logistic regression, decision trees, support vector machine (SVM) and a neural network (NN). The NN showed the best performance (AUC $\approx$ 0.80). Not raw accuracy but models based on AUC and cumulative lift were compared with a realistic evaluation using a rolling window.
Preprocessing techniques	The data spanned the years 2008–2013, and began with ~150 candidate features (client, product and socio-economic) which were thinned to 22 relevant features by a semi-automatic feature-selection procedure and then categorical encoded.
Dataset used	Real telemarketing data from a Portuguese retail bank (the data set later released as the public UCI Bank Marketing 'bank-additional' set (41,188 contacts)).
Key takeaways	1. A data-driven propensity model dramatically improves targeting versus no model contacting the top-ranked half of clients captures most subscribers. 2. Socio-economic indicators (e.g. Euribor, employment variation) are among the strongest predictors motivating their inclusion in my feature set. 3. AUC/LIFT, not accuracy, are the right metrics for this imbalanced problem.

Related work and Dataset Description 2

Title	Bank Direct Marketing Analysis of Data Mining Techniques
Harvard reference	Elsalamony, H.A. (2014) 'Bank Direct Marketing Analysis of Data Mining Techniques', International Journal of Computer Applications, 85(7), pp. 12–22.
Link	https://doi.org/10.5120/14852-3218
Modelling	Four techniques compared: a Multilayer Perceptron Neural Network (MLPNN), Tree Augmented Naïve Bayes (TAN), logistic regression and a C5.0 decision tree. C5.0 and the MLPNN gave the best classification accuracy.
Preprocessing techniques	Used the bank-full dataset (45,211 records, 17 attributes); data cleaning, categorical handling, and evaluation via overall accuracy, sensitivity and specificity on a train/test partition.
Dataset used	UCI Bank Marketing dataset (bank-full.csv, 45,211 instances) an earlier, larger-column version of the same source.
Key takeaways	1. Tree-based (C5.0) and neural models are both strong on this task, supporting my pairing of a linear baseline with a tree ensemble. 2. Attributes such as prior-campaign outcome and contact month are consistently predictive. 3. Reporting sensitivity or specificity (not just accuracy) is essential given the class imbalance.

Related work and Dataset Description 3

Title	Using data mining for bank direct marketing: an application of the CRISP-DM methodology
Harvard reference	Moro, S., Laureano, R. and Cortez, P. (2011) 'Using data mining for bank direct marketing: an application of the CRISP-DM methodology', in Proceedings of the European Simulation and Modelling Conference (ESM'2011). EUROSIS, pp. 117–121.
Link	https://core.ac.uk/outputs/55616194/
Modelling	Compared Naïve Bayes, a decision tree and an SVM within the CRISP-DM framework; the SVM achieved the best discrimination (highest AUC).
Preprocessing techniques	Structure CRISP- DM pipeline (business understanding → data understanding → preparation → modelling → evaluation) feature selection and categorical encoding on previous version of dataset.
Dataset used	A previous version of the Portuguese bank direct-marketing dataset (the predecessor of the public UCI Bank Marketing data).
Key takeaways	1. CRISP-DM provides a consistent project structure that corresponds to my SS1 to SS2 workflow. 2. SVM is a competitive alternative to be worth nothing without future work. 3. The business value is expressed as the share of subscribers reached for a given contact budget.

Publicly available dataset for Part B

Title	UCI Bank Marketing dataset (bank-additional-full.csv)
Harvard reference	Moro, S., Rita, P. and Cortez, P. (2014) Bank Marketing. UCI Machine Learning Repository. doi:10.24432/C5K306.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	41,188 records in bank-additional-full.csv. The repository also provides a 10% sample (bank-additional.csv, 4,119) and an older bank-full.csv(45,211). This project uses bank-additional-full.csv with an 80/20 stratified split (32,950 train/8,238 test).
Number of feature columns	20 input features + 1 target. (After dropping the leaky 'duration' column and adding an engineered flag, the modelling feature set is one hot expanded to ~53 columns that is displayed in FeatureEngineering.MD)
Number of classes	2 binary classification. Target 'y' = 'yes' or 'no' (term-deposit subscription). Imbalanced: ~11.3% yes, ~88.7 no.
Target range	N/A
Origin	Real data collected from direct (telephone) marketing campaigns of a Portugues retail bank between May 2008 and November 2010, compiled by Moro, Rita and Cortez and donated to the UCI Machine Learning Repository. It is genuine operational data, not synthetically generated.